

Cristian F. López R.

(he/him/his)

September, 2026

Postdoctoral Associate

Department of Applied Mathematics

University of Colorado Boulder

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[Personal website](#)

RESEARCH INTERESTS

Weak form Scientific Machine Learning, Data-driven modeling and identification of dynamical systems, Model predictive control, Theoretical and experimental nonlinear dynamics and vibrations, Signal processing, Rotating machinery fault diagnosis.

EDUCATION

2021 – 2025	Ph.D. in Mechanical Engineering and Applied Mechanics University of Nebraska – Lincoln, Lincoln, Nebraska, USA Dissertation: “Data-driven Methods for Signal Analysis and Nonlinear System Identification” M.S. in Mechanical Engineering and Applied Mechanics Advisor: Keegan J. Moore .
2015 – 2017	M.Eng. in Thermal Power Engineering Zhejiang University, Hangzhou, Zhejiang, China Thesis: “Bearing Fault Diagnosis based on Improved Stochastic Resonance Method” Advisor: Wei Zhong .
2008 – 2013	B.S. in Mechanical Engineering Escuela Politécnica Nacional, Quito, Pichincha, Ecuador Thesis: “The Design of a Heat Exchanger for the Cupola Furnace” Advisor: Fausto Oviedo .

PROFESSIONAL EMPLOYMENT

2025 – date	Postdoctoral Associate , Department of Applied Mathematics, University of Colorado Boulder, Boulder, Colorado, USA. Advisors: David Bortz , Stephen Becker .
2022 – 2025	Graduate Research Assistant – Fulbright Fellow , Department of Mechanical and Materials Engineering, University of Nebraska – Lincoln.
2021 – 2022	Graduate Teaching Assistant – Fulbright Fellow , Department of Mechanical and Materials Engineering, University of Nebraska – Lincoln.
2018 – 2021	Lecturer , Department of Basic Training, Escuela Politécnica Nacional.
2018	Lecturer , Department of Industrial Engineering, Universidad Tecnológica Indoamérica, Quito, Ecuador.
2015 – 2017	Graduate Research Assistant , Department of Energy, Zhejiang University.
2013 – 2014	Lecturer , School of Mechanical Engineering, Escuela Politécnica Nacional.

GRANT & FELLOWSHIPS

Awarded Research Grant

2026	DOE – Genesis: Transforming Science and Energy with AI. “ Weak-Form Agentic Digital Twins for Real-Time Control of Magnetically Confined Fusion Systems Under Sparse Observation ”. Awarded: \$750k. Role: Collaborator .
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Fellowships & Awards - University of Nebraska-Lincoln (Total: \$44k)

2025	5.	Doctor Honoris Causa, Humanismus cum Rigoro Scientifico. Awarded by the Universidad GESTALT – Mexico; Ibarra – Ecuador.
2023	4.	Travel Award, College of Engineering, University of Nebraska – Lincoln (\$500).
2022	3.	Travel Award, College of Engineering, University of Nebraska – Lincoln (\$500).
2021 – 2024	2.	Othmer Fellowship. University of Nebraska – Lincoln’s graduate studies for academic merit (\$12k).
2021 – 2023	1.	Fulbright Faculty Development Program to study in a Ph.D. program in the United States (\$31k).

Fellowships & Awards - Zhejiang University (Total: \$19k)

2017	2.	Best Presenter of the session, IEEE CSPA Conference, Penang, Malaysia.
2014 – 2018	1.	Chinese Government Scholarship. Language training and Master’s studies.

Fellowships & Awards - Escuela Politécnica Nacional (Total: \$7k)

2012 – 2013	2.	Merit-based Teaching Assistantship. Casting Laboratory
2013	1.	Academic Excellence. Top 2% in the Mechanical Engineering program.

PUBLICATIONS

Peer-Reviewed Journals

2026	12.	López C. , Partidge M., De Pascuale S., Lore J., Christlieb A.J., Becker S., Bortz D.M., “ WSINDy for Model Predictive Control with Applications to Fusion, Drones, and Chaos ”, <i>Proceedings of the Royal Society A</i> .
	11.	Ghosh S., Andrulonis A., López C. , Singh A., Moore K.J., “ Energy-Based Dual-Phased Dynamics Identification for Forced Nonlinear Vibrations ”, <i>Mechanical Systems and Signal Processing</i> .
	10.	López C. , Herzlieb J.E., Moore K.J., “ System Identification via Validation and Adaptation for Model Updating Applied to a Nonlinear Cantilever Beam ”, <i>ASME Journal of Computational and Nonlinear Dynamics</i> .
	9.	López C. , Moore K.J., “ System Identification via Validation and Adaptation (SIVA) for Model Identification of Nonlinear Vibrating Structures ”, <i>Nonlinear Dynamics</i> .
2025	8.	López C. , Naranjo Á., Salazar D., Moore K.J., “ Weak-form modified sparse identification of nonlinear dynamics ”, <i>Journal of Computational Physics</i> .
	7.	López C. , Moore K.J., “ Energy-based dual-phase dynamics identification of clearance nonlinearities ”, <i>Nonlinear Dynamics</i> .
	6.	López C. , Singh A., Naranjo Á., Moore K.J., “ A data-driven, energy-based approach for identifying governing dynamics in vibrating structures directly from measurements ”, <i>Mechanical Systems and Signal Processing</i> .
2024	5.	López C. , Moore K.J., “ Enhanced adaptive linear chirplet transform for crossing frequency trajectories ”, <i>Journal of Sound and Vibration</i> .
2022	4.	López C. , Naranjo A., Lu S., Moore K.J., “ Hidden Markov model based stochastic resonance and its application to bearing fault diagnosis ”, <i>Journal of Sound and Vibration</i> .
	3.	López C. , Wang D., Naranjo A., Moore K.J., “ Box-Cox-sparse-measures-based blind filtering: Understanding the difference between the maximum kurtosis deconvolution and the minimum entropy deconvolution ”, <i>Mechanical Systems and Signal Processing</i> .
2017	2.	López C. , Zhong W., Lu S., Cong F., Cortese I., “ Stochastic resonance in an underdamped system with FitzHug-Nagumo potential for weak signal detection ”, <i>Journal of Sound and Vibration</i> .

1. Zhong W., Chen X., Zhou Y., Wu Y., **López C.**, “Optimization of a solar aided coal-fired combined heat and power plant based on changeable integrate mode under different solar irradiance”, *Solar Energy*.

Preprint

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| S | Ghosh S., López C. , Singh A., Moore K.J., “Decomposition-based Energy-based Dual-Phase Dynamics Identification for Nonlinear MDOF Systems”, <i>Mechanical Systems and Signal Processing</i> . |
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CONFERENCE PRESENTATIONS & PROCEEDINGS

* indicates I presented; otherwise, a co-author presented.

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| 2027 | 17. Ghosh S., López C. , Singh A., Moore K.J. “Energy-based Dual-phase Dynamics for Nonlinear System Identification of Multi-Degree-of-Freedom Systems”, <i>International Modal Analysis Conference (IMAC) XLV</i> , Kissimmee, FL, United States, February 1-4. |
| 2026 | 16. Ghosh S., López C. , Singh A., Moore K.J. “Energy-based Dual-Phase Dynamics Identification For Multiple-Degree-of-Freedom Nonlinear Structures”, <i>NODYCON – The Fifth International Nonlinear Dynamics Conference</i> , Rome, Italy, September 20–23. |
| | 15. López C.* , Becker S., Bortz D.M. “Data-Driven MPC via WSINDy for Plasma, Drone Control, and Chaotic Systems”, <i>Society for Industrial and Applied Mathematics – Annual Meeting</i> , Cleveland, OH, United States, July 6–10. |
| | 14. López C. , Moore K.J. “Data-driven Structural System Identification via Validation and Adaptation”, <i>IMAC XLIV</i> , Palm Springs, CA, United States, January 19-22. |
| | 13. Ghosh S., Andrulonis A., López C. , Moore K.J. “A Data-driven Method for System Identification of Forced Nonlinear Vibrations Using Energy Conservation”, <i>IMAC XLIV</i> , Palm Springs, CA, United States, January 19-22. |
| 2025 | 12. López C. , Moore K.J. “Generative Adversarial Networks with Embedded Physical Knowledge for Structural System Identification”, <i>International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC/CIE)</i> , Anaheim, CA, United States, August 17–20. |
| | 11. Ghosh S., Andrulonis A., López C. , Moore K.J. “Data-driven Identification of Forced Nonlinear Vibrations using Conservation of Energy”, <i>IDETC/CIE</i> , Anaheim, CA, United States, August 17–20. |
| | 10. López C. , Singh A., Naranjo A., Moore K.J. “A Data-Driven Methodology to Identify the Governing Dynamics in Vibrating Structures Through the Analysis of Mechanical Energy”, <i>IMAC XLIII</i> , Orlando, FL, United States, February 10-13. |
| 2024 | 9. López C.* , Singh A., Naranjo A., Moore K.J. “A data-driven methodology to identify the governing dynamics in vibrating structures through the analysis of mechanical energy”, <i>European Nonlinear Oscillations Conference</i> , Delft, Netherlands, July 22-26. |
| | 8. Oliveira F., Johnson B., Singh A., Moore K.J. “Mathematical Modeling of the Nonlinear Dynamics of Bolted Joint Loosening”. I presented my colleagues’ work . <i>European Nonlinear Oscillations Conference 2024</i> , Delft, Netherlands, July 22-26. |
| | 7. López C. , Ramsey T., Moore K.J. “Digital Engineering Test and Modeling Process: Autonomous Methods for Reconciling Test and Model Results”, <i>JANNAF 18th Modeling and Simulation</i> , Oklahoma City, OK, United States, May 6–9. |
| | 6. López C.* , Moore K.J. “Enhanced Adaptive Linear Chirplet Transform for Multi-Component Signals with Intersecting Instantaneous Frequencies”, <i>IMAC XLII</i> , Orlando, FL, United States, January 29 – February 1. |
| | 5. Arroyo J., Ramsey T., López C.* , Moore K.J. “Learning Missing Data in Measurements using Meta-Modeling Techniques”, <i>IMAC XLII</i> , Orlando, FL, United States, January 29 – February 1. |
| 2023 | 4. López C.* , Singh A., Naranjo A., Moore K.J. “Data-driven Euler-Lagrange approach for Time Series Analysis”, <i>IMAC XLI</i> , Austin, TX, United States, February 13–16. |

2022	3.	López C.* , Naranjo A., Moore K.J. “Hidden Markov Model based Stochastic Resonance for bearing fault diagnosis”, <i>IMAC XL</i> , Orlando, FL, United States, February 7–10.
2017	2.	López C.* , Zhong W., Cong F., Hidalgo V., “ Bearing fault diagnosis based on Ensemble Empirical Mode Decomposition and Teager energy operator ”, <i>IEEE 13th International CSPA</i> , Penang, Malaysia, March 10–12.
2016	1.	López C.* , Zhong W., Zheng M., “ Short-term Electric Load Forecasting Based on Wavelet Neural Network, Particle Swarm Optimization and Ensemble Empirical Mode Decomposition ”, <i>Energy Procedia. 8th International Conference on Applied Energy</i> , Beijing, China, October 8–11.

WORKSHOPS

2026	4.	Scientific Machine Learning. University of Notre Dame, South Bend, IN, USA. September 26–27. Role: Attendee .
	3.	Society of Industrial and Applied Mathematics Annual Meeting. Weak-Form Scientific Machine Learning: Theory and Applications Minisymposium. Cleveland, OH, USA. July 6–10. Role: Co-organizer .
	2.	WSINDy for model predictive control with applications to plasma boundary dynamics, drone tracking trajectory, and chaotic systems. Postdoctoral seminar, University of Colorado Boulder, Boulder, CO, USA. April 29. Role: Presenter .
2025	1.	Center for Hierarchical and Robust Modeling of Non-Equilibrium Transport Annual Meeting. University of Washington, Seattle, WA, USA. December 1–3. Role: Presenter .

INDUSTRY EXPERIENCE

2022 – 2023	Cruze Distribution LLC dba ARYSE under AFOSR STTR Phase II, “STTR: Performance Knee Brace”. I used finite element models to simulate and experimentally test components of a new type of knee brace.
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TEACHING EXPERIENCE & ACTIVITIES

University of Nebraska – Lincoln

Professional Development

2022 – 2023	College of Engineering Graduate Student Teaching Fellow, and Center for the Integration of Research, Teaching, and Learning @ Nebraska Associate Level Certification. I was selected for a one-year STEM teaching program with evidence-based pedagogy and mentored course design.
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Course Taught

2021 – 2022	MENG 3250 Mechanics of Elastic Bodies (Teaching Assistant).
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Escuela Politécnica Nacional

Courses Taught (in Spanish)

2018 – 2021	- BAS030 Fundamentals of Physics - FIS145 Newtonian Mechanics: Practical Component and FIS151 Laboratory.
2013 – 2014	- MEC1B2 Casting Technology - MEC4E3 Numerical Methods - MEC4B4 Mechanics of Materials - MEC7F4 Welding Production and Maintenance.
2012 – 2013	- MEC1B2 Casting Laboratory (Laboratory Assistant and Auxiliar).

Universidad Tecnológica Indoamérica

Courses Taught (in Spanish)

2018 | - CAD/CAM/CIM and Maintenance.

Mentoring

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| 2026 | 8. | Tony Samour, University of Colorado Boulder, Master student. |
| 2025 | 7. | Jackson E. Herzlieb, Georgia Institute of Technology, Undergraduate student. |
| 2024 | 6. | Alexis Andrulonis, Georgia Institute of Technology, Undergraduate student. |
| | 5. | Sayantana Ghosh, University of Nebraska – Lincoln, Ph.D. student. |
| 2023 | 4. | Thomas Ramsey, University of Nebraska – Lincoln, Undergraduate student. |
| 2022 | 3. | Javier Arroyo, University of Nebraska – Lincoln, Ph.D. student. |
| 2014 | 2. | Sebastián Camacho. “Design and simulation of a clay modeling lathe”, Escuela Politécnica Nacional, Undergraduate student, committee member (in Spanish). |
| | 1. | Maritza Chuquirima and Lenin Cortez. “Study and Obtaining of Iron Metal from Ferruginous Sands”, Escuela Politécnica Nacional, Undergraduate students, committee member (in Spanish). |

SERVICE

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| 2024 | 4. | Certified Peer Reviewer: Elsevier, Researcher Academy. I completed the modules of the training program. |
| 2023 | 3. | Selection Committee – Fulbright Commission in Ecuador: Evaluation of applications. |
| 2022 | 2. | Intermediary for Academic Collaboration: Facilitated research partnership between University of Nebraska – Lincoln’s Mechanical & Materials Engineering Department and Escuela Politécnica Nacional’s Mechanical Engineering School. |
| 2021 – date | 1. | Reviewer:
<i>Journals</i>
– Applied Mathematical Modelling
– Chaos, Solitons and Fractals
– Communications Physics
– Discover Applied Science
– IEEE Trans. on Industrial Electronics
– IEEE Trans. on Signal Processing
– Journal of Aircraft
– Structural Health Monitoring
– Journal of Nondestructive Evaluation
– Journal of Signal, Image and Video Proc.
– Journal of Sound and Vibration
– Mechanical Systems and Signal Proc.
– Nonlinear Dynamics
– ROBOMECH Journal
– Scientific Reports
<i>Conference</i>
International Design Engineering Technical Conferences & Computers and Information in Engineering Conference |

LANGUAGES

Spanish (native), English (advanced), Mandarin (limited working proficiency).

SOFTWARE PACKAGES

A more detailed list is available at cristianlopez85.github.io/research/.

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| 2026 | 5. | WSINDY-MPC, written in MATLAB. A framework that uses WSINDYc and Model Predictive Control to identify interpretable nonlinear dynamical models from noisy measurements and use them for prediction and closed-loop control. |
| 2025 | 4. | WmSINDy, written in Python. A system identification method that identifies parsimonious nonlinear dynamical models from noisy measurements while simultaneously characterizing measurement noise. |

2025	3.	EDDI, written in MATLAB. An energy-based method for the discovery of governing equations directly from free-response measurements.
2024	2.	EALCT, written in MATLAB. A signal decomposition method for signals that have crossing instantaneous-frequency trajectories
2017	1.	SRFHN, written in MATLAB. A stochastic resonance method using an underdamped FitzHugh–Nagumo potential to enhance weak periodic signals buried in noise.